

General Chemistry Complete Study Notes

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However, errors may exist. If you find an error - whether factual, typographical, or anything that seems incorrect - please email me so I can correct it for future versions.

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Thank you for helping improve this resource!

1: The Atom & Subatomic Particles

Atom: Basic unit of an element. Molecule: Multiple atoms bonded.

Subatomic Particles:

Proton (p⁺): Charge +1, mass 1 amu. Located in nucleus. Determines element (Atomic Number, Z).

Neutron (n⁰): Charge 0, mass 1 amu. Located in nucleus. Isotope variation.

Electron (e⁻): Charge -1, mass ~0 amu (1/1836 of p⁺). Located in orbitals.

Key Terms:

Atomic Number (Z): Number of protons. (e.g., F always has 9).

Mass Number (A): Protons + Neutrons (e.g., Carbon-13 has 6p⁺, 7n⁰).

Isotope: Same Z, different A (different number neutrons). Tip: Used as tracers to track reaction pathways.

Atomic Weight: Average mass of all isotopes (weighted by abundance). Atomic weight = $m_1p_1 + m_2p_2 + \dots$

Example: Cl is 35.5 amu because Cl-35 (75%) and Cl-37 (25%) average out.

2: The Periodic Table & Organization

Groups (Columns): Same valence electron configuration → similar chemical properties.

Group 1 (Alkali Metals): Oxidation State +1, very reactive.

Group 2 (Alkaline Earth): Oxidation State +2.

Group 16 (Chalcogens): Oxidation State -2.

Group 17 (Halogens): Oxidation State -1.

Group 18 (Noble Gases): Full octet → inert.

Periods (Rows): Same principal quantum number (n).

Metal vs. Nonmetal:

Metals: Shiny, conductive, malleable, ductile.

Nonmetals: Brittle, insulators.

Metalloids: Properties of both (B, Si, Ge, As, Sb, Te).

3: Periodic Trends (Must Memorize)

Effective Nuclear Charge (Z_{eff}): Net pull on valence electrons. Increases Left $\rightarrow$ Right.

Atomic Radius: Size of atom. Decreases Left $\rightarrow$ Right, Increases Top $\rightarrow$ Bottom.

Why? More protons pull electrons in tighter.

Ionic Radius:

Cations (+): Smaller than parent atom (lost shell or increased pull).

Anions (-): Larger than parent atom (extra repulsion among electrons).

Ionization Energy (IE): Energy to remove an electron. Increases Left $\rightarrow$ Right, Decreases Top $\rightarrow$ Bottom.

Exception: IE2 is always $>$ IE1 (even for Ca going to octet).

Electron Affinity (EA): Energy released when atom gains electron. Highest for Cl (not F).

Why Cl $>$ F? F is too small; adding electron causes high electron-electron repulsion.

Electronegativity (EN): Pull on shared electrons. Most EN is Fluorine (4.0). Increases Left $\rightarrow$ Right, Bottom $\rightarrow$ Top.

4: Quantum Numbers (Coordinates for Electrons)

Principal (n): Energy level/shell (1, 2, 3...). Higher n = higher energy.

Azimuthal (l): Subshell shape. (0=s, 1=p, 2=d, 3=f).

s: 1 orbital, 2 electrons | p: 3 orbitals, 6 electrons | d: 5 orbitals, 10 electrons | f: 7 orbitals, 14 electrons.

Magnetic (m_l): Orbital orientation (- l to + l).

p: -1, 0, +1 | d: -2, -1, 0, +1, +2.

Spin (m_s): +1/2 (up) or -1/2 (down).

Example Fluorine (9 electrons): 1s², 2s², 2p⁵. One electron might be: $n=2$, $l=1$, $m_l=0$, $m_s=+1/2$.

5: Electron Configuration & Exceptions

Order of Filling: 1s, 2s, 2p, 3s, 3p, 4s (fills before 3d), 3d, 4p, 5s, 4d...

Noble Gas Shorthand: [Ne] 3s² 3p⁵ (Chlorine).

The "Weird" Transition Metals (Must Know):

Chromium (Cr): Expected [Ar] 4s² 3d⁴ $\rightarrow$ Actual [Ar] 4s¹ 3d⁵ (Half-filled d = stable).

Copper (Cu): Expected [Ar] 4s² 3d⁹ $\rightarrow$ Actual [Ar] 4s¹ 3d¹⁰ (Filled d = stable).

Ions: Remove electrons from highest n first.

Trick: Fe²⁺ removes 4s electron before 3d. Configuration = [Ar] 3d⁶ (not 4s² 3d⁴).

6: VSEPR Theory & Molecular Geometry

Goal: Minimize electron pair repulsion. Electron Groups = Bonds (single/double/triple count as 1 group) + Lone Pairs.

2 Electron Groups: Orbital Geometry = Linear, Molecular Geometry = Linear, Bond Angle = 180° , Example: CO_2

3 Electron Groups: Orbital Geometry = Trigonal Planar, Molecular Geometry = Trigonal Planar (no lone pairs) angle 120° (BF_3) or Bent (1 lone pair) angle $<120^\circ$ (SO_2)

4 Electron Groups: Orbital Geometry = Tetrahedral, Molecular Geometry = Tetrahedral (no lone pairs) angle 109.5° (CH_4) or Trigonal Pyramidal (1 lone pair) angle $<109.5^\circ$ (NH_3) or Bent (2 lone pairs) angle $<109.5^\circ$ (H_2O)

7: Bonding Fundamentals

Ionic Bond: Transfer of electrons. EN difference > 1.7 .

Strength: Very strong electrostatic attraction. High melting point. Dissolves in water (electrolyte).

Note: Covalent bonds (especially triple bonds like $\text{N}\equiv\text{N}$) are generally stronger *within a molecule* than ionic bonds. Ionic bonds are strong between ions in a crystal lattice.

Covalent Bond: Sharing electrons. EN difference $0.5 - 1.7$ (polar) or <0.5 (nonpolar).

Sigma (σ) Bond: Head-on overlap. First bond is always σ . Strong.

Pi (π) Bond: Sideways overlap. Second/third bonds are π . Weaker than σ .

Bond Order: Single (1σ), Double ($1\sigma + 1\pi$), Triple ($1\sigma + 2\pi$).

Higher order = Shorter bond length = Stronger bond.

8: Intermolecular Forces (IMFs) - Ranked Strongest to Weakest

1. Ion-Dipole: Ion + polar molecule. (e.g., Na^+ in H_2O). Very strong.
2. Hydrogen Bonding: H bonded to F, O, or N. (e.g., H_2O , DNA base pairs). Check: Must be H-F, H-O, H-N.
3. Dipole-Dipole: Polar molecule + Polar molecule.
4. London Dispersion (LDF): Temporary dipole in nonpolar molecules. Trend: Increases with molecular weight (more electrons, more polarizable).

Relation to Properties:

High BP/MP = Strong IMFs.

Low Vapor Pressure = Strong IMFs.

9: Types of Chemical Reactions

Combustion: $\text{Fuel} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$. Balancing Tip: $\text{C} \rightarrow \text{H} \rightarrow \text{O}$.

Single Displacement: $\text{A} + \text{BC} \rightarrow \text{AC} + \text{B}$.

Double Displacement: $\text{AB} + \text{CD} \rightarrow \text{AD} + \text{CB}$. (Often forms precipitate or gas).

Redox (Oxidation-Reduction): Change in oxidation states.

Organic trick: OIL RIG (Oxidation Is Loss, Reduction Is Gain).

Oxidation = Gains O or loses H. Reduction = Gains H or loses O.

Acid-Base: Transfer of H^+ (proton).

10: Equilibrium Constant (K_{eq})

Definition: Ratio of $[\text{Products}]/[\text{Reactants}]$ at equilibrium. No units (usually).

$K_{\text{eq}} = \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b}$ for $a\text{A} + b\text{B} \rightleftharpoons c\text{C} + d\text{D}$.

Rules:

Omit solids & liquids. Only include (aq) and (g).

Raise to stoichiometric coefficient.

What K tells you:

$K \gg 1$: Products favored, spontaneous forward ($\Delta G < 0$).

$K \ll 1$: Reactants favored, NOT spontaneous ($\Delta G > 0$).

$K = 1$: Equal amounts, $\Delta G = 0$.

Reaction Quotient (Q): Same formula, but NOT at equilibrium.

If $Q < K \rightarrow$ Forward ($\rightarrow$)

If $Q > K \rightarrow$ Reverse ($\leftarrow$)

11: Le Châtelier's Principle (Shifts)

If you stress a system, it shifts to relieve the stress.

Concentration: Add reactant $\rightarrow$ Shifts right. Remove product $\rightarrow$ Shifts right.

Pressure/Volume (Gas only):

Increase P (or decrease V) $\rightarrow$ Shifts toward side with fewer gas moles.

Decrease P (or increase V) $\rightarrow$ Shifts toward side with more gas moles.

Temperature:

Heat + Endothermic ($\Delta H > 0$) $\rightarrow$ Adding heat shifts Right.

Heat + Exothermic ($\Delta H < 0$) $\rightarrow$ Adding heat shifts Left.

Catalyst: Speeds up forward AND reverse equally. Does NOT shift equilibrium. Lowers E_a .

12: Kinetics vs. Thermodynamics (Crucial distinction)

Kinetics: Speed of reaction. Kinetically favored = Low Activation Energy (E_a).

Control: Temperature, Catalyst, Concentration (except 0th order).

Thermodynamics: Favorability/Stability. Thermodynamically favored = Negative ΔG (Exergonic).

The Graph:

Reactants $\rightarrow$ Transition State ($\Delta G^\ddagger$) $\rightarrow$ Products.

ΔG (overall) = Products - Reactants (Thermo).

E_a = Transition State - Reactants (Kinetic).

Trap: A reaction can be thermodynamically favorable ($\Delta G < 0$) but kinetically slow (high E_a). (e.g., Diamond turning to graphite).

13: Reaction Rates & Rate Laws

Rate Law: $\text{rate} = k [A]^x [B]^y$. (Determined experimentally OR by the slow step).

Order (x or y):

0: [A] doubles $\rightarrow$ rate unchanged. (k units = M/s).

1: [A] doubles $\rightarrow$ rate doubles. (k units = s^{-1}).

2: [A] doubles $\rightarrow$ rate quadruples. (k units = $M^{-1} s^{-1}$).

Finding Order from Data: Find two trials where only [A] changes. See how rate changes.

Slow Step (Rate Determining Step): The rate law is the product of the reactants in the slow step (raised to their coefficients).

Example: Step 2 (slow): $N_2O_2 + H_2 \rightarrow \dots$ Rate = $k[N_2O_2][H_2]$.

14: Integrated Rate Laws (Graphing)

0th Order: [A] vs. time. Straight line, slope = -k.

1st Order: $\ln[A]$ vs. time. Straight line, slope = -k.

Half-life ($t_{1/2}$) = $0.693/k$ (Constant for 1st order).

2nd Order: $1/[A]$ vs. time. Straight line, slope = +k.

Shortcut: If they give you an equation with " $\ln$ ", it's 1st order. If it has " $1/[A]$ ", it's 2nd order.

15: Activation Energy & Catalysts

Arrhenius Equation: $k = Ae^{(-E_a/RT)}$

k = rate constant, A = collision frequency, E_a = activation energy.

To increase k (speed up):

Increase T (molecules move faster).

Decrease E_a (add catalyst).

Increase A (proper orientation – enzymes do this).

Catalysts:

Homogeneous: Same phase as reactants.

Heterogeneous: Different phase (e.g., solid catalyst in liquid).

Enzymes: Biological catalysts (proteins on). Do NOT change ΔG or K.

16: Thermodynamics Basics (ΔH , ΔS , ΔG)

System Types:

Isolated: Nothing exchanged (matter or energy).

Closed: Energy exchanged, matter NOT exchanged.

Open: Both matter and energy exchanged.

State Functions: Path independent (P, V, T, H, S, G).

Path Functions: Q (heat), W (work) — these depend on the pathway taken.

Enthalpy (ΔH): Heat of reaction at constant pressure.

Exothermic ($\Delta H < 0$): Releases heat (feels hot). Bonds formed > bonds broken.

Endothermic ($\Delta H > 0$): Absorbs heat (feels cold). Bonds broken > bonds formed.

Entropy (ΔS): Disorder of a system.

$\Delta S > 0$: More disorder (solid $\rightarrow$ liquid, liquid $\rightarrow$ gas, more gas moles, dissolving solids).

$\Delta S < 0$: Less disorder (gas $\rightarrow$ liquid, gas $\rightarrow$ solid, condensation, freezing).

17: Gibbs Free Energy (The Decider)

Equation: $\Delta G = \Delta H - T\Delta S$.

Spontaneity (Exergonic): $\Delta G < 0$.

Four Scenarios:

1. ΔH (-), ΔS (+): $\Delta G < 0$ at All T. (Always spontaneous).
2. ΔH (+), ΔS (-): $\Delta G > 0$ at All T. (Never spontaneous).
3. ΔH (-), ΔS (-): $\Delta G < 0$ at Low T.
4. ΔH (+), ΔS (+): $\Delta G < 0$ at High T.

$\Delta G^\circ = -RT \ln(K_{eq})$. (Links thermo to equilibrium).

If ΔG° is negative, $K > 1$.

If ΔG° is positive, $K < 1$.

18: Phase Changes & Phase Diagrams

Endothermic (absorb heat, ΔH +, ΔS +): Melting (S $\rightarrow$ L), Vaporization (L $\rightarrow$ G), Sublimation (S $\rightarrow$ G).

Exothermic (release heat, ΔH -, ΔS -): Freezing (L $\rightarrow$ S), Condensation (G $\rightarrow$ L), Deposition (G $\rightarrow$ S).

Phase Diagram (P vs. T):

Triple Point: All 3 phases coexist.

Critical Point: End of liquid-gas line. Beyond it = Supercritical fluid.

Heating Curve (T vs. Heat):

Slopes: $Q = mC\Delta T$ (change T).

Plateaus: $Q = mL$ (phase change, constant T).

19: Gases & Ideal Gas Law

Ideal Gas Law: $PV = nRT$.

$R = 0.0821 \text{ L}\cdot\text{atm} / (\text{mol}\cdot\text{K})$.

STP (Standard Temp & Pressure): 0°C (273K), 1 atm. 1 mole = 22.4 L.

Ideal Gas Assumptions: No volume, no IMFs. (True at High T, Low P).

Kinetic Molecular Theory (KMT):

$KE_{\text{avg}} = (3/2)RT$. Temp is average KE.

Lighter gasses move faster (Graham's Law).

Dalton's Law (Partial Pressure): $P_{\text{total}} = P_1 + P_2 + P_3$.

$P_{\text{partial}} = (\text{mole fraction}) \times P_{\text{total}}$.

20: Real Gases & Deviations

Real Gases (Low T, High P): Attractions dominate $\rightarrow$ Pressure is lower than ideal. Molecules stick together slightly.

Very High P: Volume of molecules dominates $\rightarrow$ Volume is higher than ideal.

Van der Waals Equation: $(P + a(n/V)^2)(V - nb) = nRT$.

a corrects for attraction. b corrects for volume.

21: Solutions & Concentration Units

Molarity (M): mol solute / L solution. (Temp dependent – volume changes).

Molality (m): mol solute / kg solvent. (Temp independent). Used for colligative properties.

Mole Fraction (X): mol A / total mol.

Dilution: $M_1V_1 = M_2V_2$. (Moles are conserved).

Solubility Rules (NAG SAG are soluble):

Nitrates, Acetates, Group 1, Sulfates, Ammonium, Group 17 (halides).

Exceptions: PMS (Pb, Hg, Ag) and Castro Bear (Ca, Sr, Ba) are insoluble with halides/sulfates.

22: Colligative Properties (Depends on number of particles)

Van't Hoff Factor (i): Number of particles after dissociation. ($\text{NaCl} = 2$, $\text{CaCl}_2 = 3$).

Vapor Pressure Lowering: $\Delta P = i \times X_{\text{solute}} \times P_{\text{solvent}}$. (Raoult's Law).

Boiling Point Elevation: $\Delta T_b = i \times K_b \times m$.

Freezing Point Depression: $\Delta T_f = i \times K_f \times m$.

Osmotic Pressure (Π): $\Pi = i \times M \times R \times T$.

Tip: Adding salt to water raises BP and lowers FP.

23: Acids & Bases (Definitions)

Arrhenius: Acid = H^+ producer (HCl). Base = OH^- producer (NaOH).

Brønsted-Lowry: Acid = H^+ donor. Base = H^+ acceptor. (NH_3 is a base).

Lewis (Most Broad): Acid = electron pair acceptor. Base = electron pair donor. (BH_3 is acid).

Amphoteric: Can be acid or base (H_2O , HCO_3^-).

Conjugates: $HA \rightarrow A^-$ (conj. base). $A^- + H^+ \rightarrow HA$ (conj. acid).

Strong Acid $\rightarrow$ Weak Conj. Base (Cl^- is stable).

Weak Acid $\rightarrow$ Strong Conj. Base (F^- wants H^+).

24: pH, pOH, and pKa

$pH = -\log[H^+]$. (If $[H^+] = 1 \times 10^{-5}$, $pH = 5$).

$pOH = -\log[OH^-]$.

$pH + pOH = 14$ (at $25^\circ C$).

Strong Acids (Memorize): HCl, HBr, HI, HNO_3 , H_2SO_4 , $HClO_4$.

Strong Bases (Memorize): LiOH, NaOH, KOH, $Ca(OH)_2$, $Sr(OH)_2$, $Ba(OH)_2$.

$pKa = -\log(Ka)$. Low pKa = Strong Acid.

$K_w = 1e-14 = [H^+][OH^-] = K_a \times K_b$.

$pKa + pKb = 14$.

25: Weak Acids & Henderson-Hasselbalch

HH Equation: $pH = pKa + \log([A^-]/[HA])$.

Use ONLY for buffers (weak acid + conjugate salt).

Half-Equivalence Point: $pH = pKa$. ($[A^-] = [HA]$).

Log Shortcut:

If $[A^-] = [HA] \rightarrow \log(1)=0 \rightarrow pH = pKa$.

If $[A^-] > [HA] \rightarrow pH > pKa$ (basic).

If $[A^-] < [HA] \rightarrow pH < pKa$ (acidic).

10x difference $\rightarrow pH = pKa \pm 1$.

26: Buffers & Titrations

Buffer: Resists pH change. Needs weak acid/base + its conjugate.

Buffer Capacity: Max within ± 1 pH of pKa.

Titration Curve (Strong Base into Weak Acid):

Start: Low pH (weak acid only).

Half-equivalence: pH = pKa, slope is flat (buffer zone).

Equivalence Point: pH > 7 (because conjugate base hydrolyzes). Moles acid = moles base.

Beyond: pH determined by excess strong base.

Polyprotic Acid Curve: Multiple "steps" (buffers), multiple equivalence points.

27: Solubility Equilibrium (Ksp)

$K_{sp} = [\text{cation}]^m [\text{anion}]^n$ (Solid omitted — solids never appear in K_{sp}).

Example: $\text{PbCl}_2(s) \rightleftharpoons \text{Pb}^{2+}(aq) + 2\text{Cl}^-(aq)$. $K_{sp} = [\text{Pb}^{2+}][\text{Cl}^-]^2$

Molar Solubility (s): If $s = [\text{Pb}^{2+}]$, then $[\text{Cl}^-] = 2s$. $K_{sp} = (s)(2s)^2 = 4s^3$.

Common Ion Effect: Add Cl^- (from NaCl). $Q > K_{sp} \rightarrow$ Less soluble (shifts left toward solid).

Predicting Precipitation: If $Q > K_{sp} \rightarrow$ precipitate forms. If $Q < K_{sp} \rightarrow$ more can dissolve.

28: Redox Fundamentals (Oxidation States)

Rules for Assigning Oxidation States:

1. Free element = 0 (O_2 , Na).
2. Sum = total charge of ion/molecule.
3. Group 1 = +1, Group 2 = +2.
4. F = -1, H = +1 (except metal hydrides = -1), O = -2 (except peroxides = -1).

Oxidizing Agent: Gets reduced (gains electrons). (e.g., O_2 , KMnO_4).

Reducing Agent: Gets oxidized (loses electrons). (e.g., NADH, Zn metal).

29: Balancing Redox (Half-Reaction Method)

Acidic Solution:

1. Write half-reactions.
2. Balance atoms (except H/O).
3. Balance O with H_2O .
4. Balance H with H^+ .
5. Balance charge with electrons.
6. Multiply to match electrons and add.

Basic Solution: Same steps, but AFTER balancing in acidic, add OH^- to both sides to neutralize H^+ into H_2O .

Example (Acidic): $\text{MnO}_4^- + \text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + \text{Fe}^{3+}$

Reduction: $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$.

Oxidation: $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + 1\text{e}^-$ (x5).

Final: $\text{MnO}_4^- + 8\text{H}^+ + 5\text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O} + 5\text{Fe}^{3+}$.

30: Electrochemical Cells (Galvanic vs. Electrolytic)

Galvanic (Voltaic):

Spontaneous ($\Delta G < 0$, $E^\circ_{\text{cell}} > 0$).

Anode (-) : Oxidation (loses mass).

Cathode (+) : Reduction (gains mass).

Salt bridge maintains neutrality.

Electrolytic:

Nonspontaneous ($\Delta G > 0$, $E^\circ_{\text{cell}} < 0$). Requires battery.

Anode (+) : Oxidation (forced).

Cathode (-) : Reduction (forced).

Memory Trick: An Ox (Anode Oxidation), Red Cat (Reduction Cathode).

31: Cell Potential & Thermodynamics

$E^\circ_{\text{cell}} = E^\circ(\text{cathode}) - E^\circ(\text{anode})$. (Use reduction potentials from table).

$\Delta G^\circ = -nFE^\circ_{\text{cell}}$. (n = mol electrons, $F = 96485 \text{ C/mol}$).

Nernst Equation: $E = E^\circ - (0.0592/n) \log(Q)$ at 298K.

As battery runs, Q increases, E decreases until $E=0$ (dead).

Concentration Cell: Same electrodes, different [ions]. Spontaneous until concentrations equalize.

32: Electrolysis & Faraday's Law

Electrolysis: Using electricity to drive nonspontaneous redox (electrolytic cell).

Faraday's Law: Amount of substance produced is proportional to charge.

Current (I) = Charge (Q) / Time (t) $\rightarrow I = Q/t$

Charge (Coulombs) = Current (Amps) $\times$ Time (seconds) $\rightarrow Q = I \times t$

Moles of electrons = $(I \times t) / F$ where $F = 96485 \text{ C/mol e}^-$

Stoichiometry Trick:

1. $I \times t = \text{Coulombs (Q)}$
2. $\text{Coulombs} / F = \text{moles of electrons transferred}$
3. $\text{Moles e}^- \rightarrow \text{moles of metal (using balanced half-reaction coefficients)}$
4. $\text{Moles metal} \rightarrow \text{grams metal (multiply by molecular weight)}$

33: Spectator Ions & Net Ionic Equations

Complete Ionic Equation: All soluble strong electrolytes split into ions.

Net Ionic Equation: Remove spectator ions (ions that appear same on both sides).

Example: $\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{NaNO}_3(\text{aq})$

Complete: $\text{Ag}^+ + \text{NO}_3^- + \text{Na}^+ + \text{Cl}^- \rightarrow \text{AgCl}(\text{s}) + \text{Na}^+ + \text{NO}_3^-$

Net Ionic: $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$.

34: Nuclear Chemistry (basics)

Alpha (α): Helium nucleus ($4/2 \text{ He}$). Low penetration. $Z=2$, $A=4$.

Beta (β^-): Electron ($0/-1 \text{ e}$). $Z+1$, A same. (Neutron $\rightarrow$ Proton).

Positron (β^+): Positive electron ($0/+1 \text{ e}$). $Z-1$, A same.

Gamma (γ): High energy photon. No change in Z or A .

Half-life ($t_{1/2}$): Time for half of isotope to decay. $t_{1/2} = \ln(2)/\lambda$.

35: Experimental Lab Techniques (For GC)

Titration Setup: Buret (titrant) $\rightarrow$ Flask (analyte) + indicator.

Indicator Choice: pK_a of indicator $\approx$ pH at equivalence point.

Spectrophotometry: Measures light absorbed. Beer-Lambert Law: $A = \epsilon bc$.

A = absorbance, ϵ = molar absorptivity, b = path length, c = concentration.

Higher concentration = Higher absorbance.

Distillation: Separates by boiling point. Lower BP vaporizes first.

Chromatography (General): Separates by polarity.

Stationary phase: Polar (silica). Mobile phase: Nonpolar (hexane).

Nonpolar compounds move faster.

36: Math Skill – Scientific Notation & Logs

Multiplying: Add exponents. ($10^5 \times 10^3 = 10^8$)

Dividing: Subtract exponents. ($10^5 / 10^3 = 10^2$)

Square Roots: Halve the exponent. $\sqrt{10^6} = 10^3$.

Log Shortcuts (pH):

If $[\text{H}^+] = 1 \times 10^{-n}$, $\text{pH} = n$.

If $[\text{H}^+] = 3 \times 10^{-5}$, $\text{pH} = 5 - \log(3) = 5 - 0.48 = 4.52$.

Anti-log: If $\text{pH} = 4.52$, $[\text{H}^+] = 3 \times 10^{-5}$.

37: Math Skill – Dimensional Analysis

The Golden Rule: Units cancel. Always write them.

Stoichiometry Roadmap:

Grams A $\rightarrow$ Moles A (divide by MW) $\rightarrow$ Moles B (multiply by mole ratio) $\rightarrow$ Grams B (multiply by MW).

Mole Ratio: Comes from balanced equation coefficients.

Example: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. 2 mol H_2 : 1 mol O_2 .

38: Light & Spectroscopy (Quantum Chem)

$E = h\nu = hc/\lambda$. (E = energy, h = Planck's, ν = frequency, λ = wavelength).

Longer λ = Lower frequency = Lower Energy.

Absorption: Electron jumps from ground $\rightarrow$ excited. (Absorbs specific λ).

Emission: Electron falls from excited $\rightarrow$ ground. (Emits specific λ).

Fluorescence: Immediate emission (lower energy/longer λ than absorbed).

Beer-Lambert Law: $A = \epsilon bc$. A directly proportional to concentration.

39: Nuclear Decay & Kinetics

Decay follows 1st order kinetics.

$N_t = N_0 \times e^{(-\lambda t)}$. (λ = decay constant).

$t_{1/2} = 0.693 / \lambda$.

Trick: Calculate fraction remaining. After 1 $t_{1/2} = 1/2$ left. After 2 $t_{1/2} = 1/4$ left. After 3 $t_{1/2} = 1/8$ left.

40: Thermodynamics – Hess's Law

$\Delta H^\circ_{\text{rxn}} = \sum \Delta H^\circ_f(\text{products}) - \sum \Delta H^\circ_f(\text{reactants})$.

Hess's Law: If you reverse a reaction, flip the sign of ΔH . If you multiply by coefficient, multiply ΔH .

Bond Enthalpy Method: $\Delta H = \sum(\text{bonds broken}) - \sum(\text{bonds formed})$.

(Positive if endothermic – needs energy to break).

41: Entropy (ΔS) Predictions

Gas > Liquid > Solid.

More molecules > Fewer molecules.

Dissolving solid: $\Delta S +$ (more disorder).

Dissolving gas: $\Delta S -$ (gas becomes trapped in liquid).

Increasing Temperature: $\Delta S +$ (more molecular motion).

42: Free Energy & Non-Standard Conditions

$\Delta G = \Delta G^\circ + RT \ln(Q)$.

Use when not 1M, 298K, 1 atm.

Relationship to Work: ΔG = Maximum useful work (non-expansion).

Coupling Reactions: Unfavorable ($\Delta G +$) can be driven by favorable ($\Delta G -$) reaction in a pathway. (e.g., ATP hydrolysis driving muscle contraction).

43: Acid Strength Trends (Binary & Oxyacids)

Binary Acids (H-X): Bond strength decreases down group $\rightarrow$ Acid strength increases down group. (HI > HBr > HCl > HF).

Oxyacids (HO-X):

More O atoms: Higher acid strength. (HClO₄ > HClO₃ > HClO₂ > HClO).

Electronegativity of X: Higher EN of X $\rightarrow$ Higher acid strength. (HOCl > HOBr > HOI).

44: Lewis Acids & Bases (Coordination Chem)

Lewis Base: Lone pair donor (ligand). (NH_3 , H_2O , CN^- , Cl^-).

Lewis Acid: Lone pair acceptor (metal center). (Fe^{3+} , Cu^{2+} , BH_3).

Complex Ions: $[\text{Fe}(\text{CN})_6]^{3-}$.

Coordination Number: Number of bonds to metal (here, 6).

Chelation: Ligand binds with multiple atoms (e.g., EDTA).

45: Electrochemistry – Nernst Equation Practice

$E = E^\circ - (0.0592/n) \log(Q)$. (At 298K).

At Equilibrium: $Q = K$, $E = 0$. So $0 = E^\circ - (0.0592/n) \log(K) \rightarrow \log(K) = nE^\circ/0.0592$.

Concentration Cell Example: Both sides Cu/Cu^{2+} . $E^\circ = 0$.

$E = -(0.0592/2) \log([\text{dilute}]/[\text{concentrated}])$.

46: Kinetics – Activation Energy Diagrams

Labeling: Reactants, Products, ΔH (Endo or Exo), E_a (forward), E_a (reverse).

$E_a(\text{reverse}) = E_a(\text{forward}) - \Delta H$.

Catalyst lowers E_a for both forward and reverse. Does NOT change ΔH .

47: Phase Diagrams – High Yield Interpretation

Solid Region: High P, Low T.

Gas Region: Low P, High T.

Liquid Region: Between solid/gas.

Sublimation Curve: Separates Solid/Gas.

Vaporization Curve: Separates Liquid/Gas.

Fusion Curve: Separates Solid/Liquid. (Negative slope for water).

Critical Temperature (T_c): Gas cannot be liquified above this temp.

48: Henry's Law & Gas Solubility

Henry's Law: $C = k \times P_{\text{gas}}$.

C = concentration dissolved, P_{gas} = partial pressure above liquid.

High pressure = High solubility.

High temperature = Low solubility for gases (exothermic dissolution).

Example: Decompression sickness ("the bends") – N_2 bubbles out of blood when pressure drops too fast.

49: Colloids & Suspensions

Solution: Homogeneous, <1 nm, does not scatter light.

Colloid: Heterogeneous, 1-1000 nm, scatters light (Tyndall effect). (e.g., milk, fog).

Suspension: >1000 nm, settles out over time. (e.g., mud in water).

50: Final Study Strategy for GC

1. Prioritize High-Yield Topics: Periodic Trends, IMFs, Equilibrium, Thermodynamics ($\Delta G = \Delta H - T\Delta S$), Redox, Kinetics.
2. Math Focus: pH and pOH calculations (logs), Nernst equation, Rate laws and units of k, Dilutions ($M_1V_1 = M_2V_2$), Ksp expressions.
3. Common Traps to Avoid:
 - Confusing K (equilibrium constant) with k (rate constant) — they are completely different.
 - Ignoring state symbols — solids and liquids NEVER go into Keq or Ksp expressions.
 - Forgetting the Van't Hoff factor (i) for colligative properties (especially with CaCl_2 , MgCl_2 , etc.).
 - Thinking $E^\circ_{\text{cell}} = 0$ for concentration cells — it is, but the actual E is NOT zero (use Nernst).
 - Assuming all exothermic reactions are spontaneous — they are NOT (ΔG also depends on ΔS and T).
 - Forgetting to convert $^\circ\text{C}$ to K in gas law and thermodynamics problems.
4. Application Over Memorization: Don't just memorize facts. Be able to explain concepts like:
 - Why HF is a weak acid but HCl is strong (bond strength, not electronegativity).
 - Why ice floats (hydrogen bonding creates open lattice $\rightarrow$ lower density).
 - Why adding salt to water raises boiling point and lowers freezing point.

End of Notes